higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 4

(12041014)

20 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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QUESTION 1

1.1 Explain the term *bonding*. (3)

1.2 What should the potential difference between objects connected to earth through conductive wires be with respect to each other? (1)

1.3 Name TWO effects that electrical accidents can have on property. (2)

1.4 Explain why we should not use water to extinguish an electrical fire. (2)

1.5 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–D) next to the question number (1.5.1–1.5.3) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.5.1	Fires that involve electrical or electronic equipment such as motors, generators and electrical wiring	A	Class A
		B	Class B
1.5.2	Fires that involve flammable or combustible metals such as magnesium, sodium, titanium, potassium, uranium, lithium, plutonium and calcium	C	Class C
		D	Class D
1.5.3	Fires that involve ordinary combustible materials such as wood, paper and textiles		

(3 × 1) (3)

1.6 Name TWO types of workplace violence. (2)

1.7 Identify SEVEN important aspects that 'The Permit to Work' document should clearly state before work can start on electrical systems. (7)

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QUESTION 2

2.1 Regulation 5.1.3 of SANS 10142-1 refers to 'earth fault current protection'.

Complete the following sentence by writing down the missing words next to the question number (a–c) in the ANSWER BOOK.

2.1.1 A new electrical (a) ... shall not be (b) ... to a supply unless the supply includes a (c) ... conductor. (3)

2.1.2 Explain the expression *electrical separation of a circuit*. (2)

2.2 Draw a neat fully labelled circuit diagram indicating how one light is controlled form three different switch locations using intermediate switching. Earthing of components must be indicated on circuit diagram as per SANS 10142-1 (8)

2.3 2.3.1 What is the energy regulator switch used in a stove also known as? (1)

2.3.2 State what this energy regulator switch is used for in a stove? (1)

2.3.3 Make a neat labelled layout diagram of the connections for this energy regulating switch used in a stove. (5)

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QUESTION 3

3.1 3.1.1 Explain the term Luminous efficiency or efficacy with reference to lamps. (2)

3.1.2 What symbol is used to identify Luminous efficiency or efficacy? (1)

3.1.3 What is the unit of measurement for Luminous efficiency or efficacy? (1)

3.2 The quality of lighting is very important as it can be harmful to viewers.

State TWO methods that can be used to avoid shadows. (2)

3.3 Explain the basic operation of a tungsten filament incandescent lamp. (3)

3.4 Tungsten halogen lamps use a tungsten filament in a gas filled, light-transmitting envelope.

List FOUR advantages of tungsten halogen lamps. (4)

3.5 3.5.1 What does the acronym CFL stand for? (1)

3.5.2 Name the TWO parts CFLs consist of. (2)

3.5.3 Name TWO types of CFL lamp fittings. (2)

3.5.4 Name TWO disadvantages of CFL lamps. (2)

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QUESTION 4

4.1 A technician can use various techniques to identify and locate the cause of faults in domestic appliances and portable power tools.

Identify FIVE of these fault-finding techniques. (5)

4.2 You have to perform circuit continuity testing on a single-phase capacitor start induction run motor used in a washing machine.

Describe FOUR tests you would perform during a circuit continuity test to locate defects. (4)

4.3 Name FOUR types of electrical flex that are used on most types of electrical domestic appliances and portable power tools. (4)

4.4 After fault-finding and repair work has been completed on any electrical domestic appliance or power tool a durable tag should be attached to the equipment.

List THREE examples of information that should be recorded on the tag to provide a history for future repairs. (3)

4.5 A power tool used in the workshop has been found to be defective.

Describe FOUR actions that should be taken to prevent that the defective tool is used again. (4)

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QUESTION 5

5.1 Electric motors are classified according to the type of enclosure they use, the conditions under which they operate, and the degree of ventilation required.

Name FOUR types of motor enclosures and give an application for each type of enclosure. (4 × 2) (8)

5.2 Name FIVE types of couplings that can be used to couple an electric motor to a mechanical load. (5)

5.3 You need to look for several features when inspecting belt drive guards.

Describe FOUR important features that a well-designed belt drive guard should have. (4)

5.4 List the THREE main parts of a three-phase induction motor. (3)

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TOTAL: 100

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